BUCKMINSTERFULLERENE IN WATER TREATMENT SYSTEMS

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INTRODUCTION

WHAT IS BUCKMINSTERFULLERENE?

Buckminsterfullerene or C60 is a spherical, hollow molecule of 60 carbon atom arranged like a soccer ball. It has a simple molecular structure with strong covalent bonds and weak intermolecular forces of attraction. Its symmetric cage-like structure makes it strong and rigid.

How can doped buckminsterfullerene support water treatment systems?

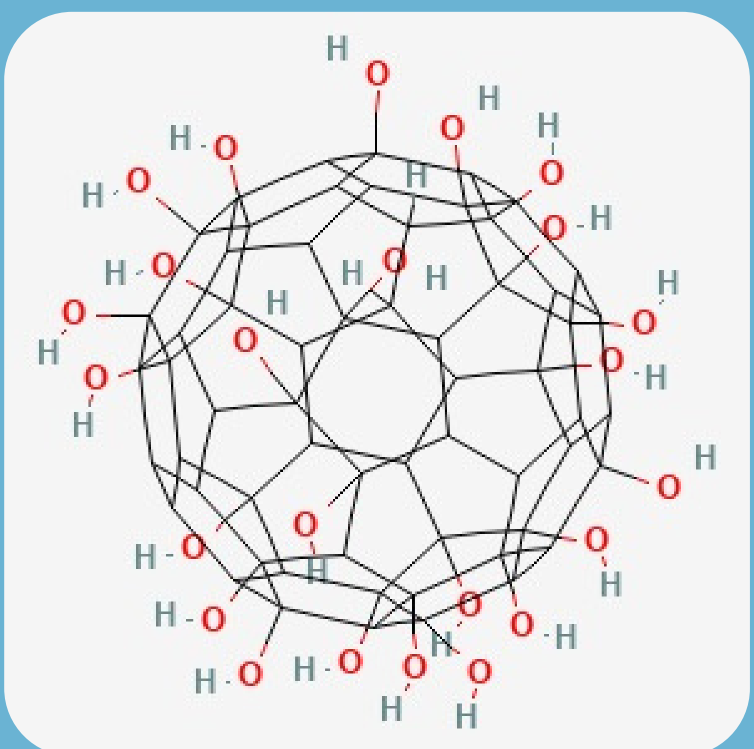
1. Being a potent antioxidant and antimicrobial itself, buckminsterfullerene brings by a many benefits to the human recovery
2. Doped buckminsterfullerene, can enhance the adsorption behaviour of fullerenes towards various gases and pollutants, giving rise to environmental monitoring
3. Specifically, C60(OH)24 can act in a water treatment system's by modifying the ultrafiltration membrane

HOW DO WE OBTAIN BUCKMINSTERFULLERENE AND ITS DOPES

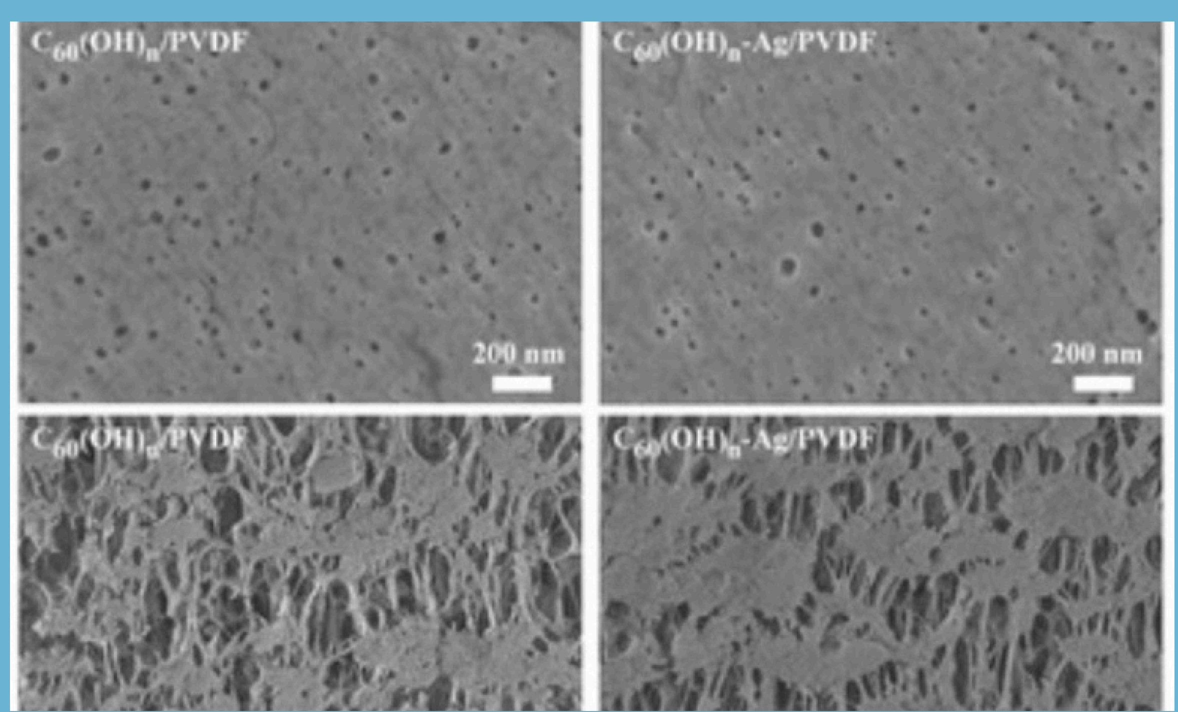
1.GetC60(buckminsterfullerene) .Graphite which is pure carbon, is heated very strongly using electricity or a laser the carbon turn into smoke. The smoke is collected and C60is taken out using a liquid and cleaned to get pure carbon.
2. Next -OH group are added to C60 to make a fullerenol. This change C60 into C60(OH)24, Which can dissolve in water.

Common Method

- Alkaline hydroxylation:
C60 reacts with NaOH or KOH and oxygen or H2O2 to add –OH groups.
- Hydrogen peroxide method:
C60 reacts with H2O2 to form water-soluble fullerenols.
- Phase-transfer method:
A catalyst helps C60 move into water so –OH groups can attach easily.



hyodoxyl fullerene



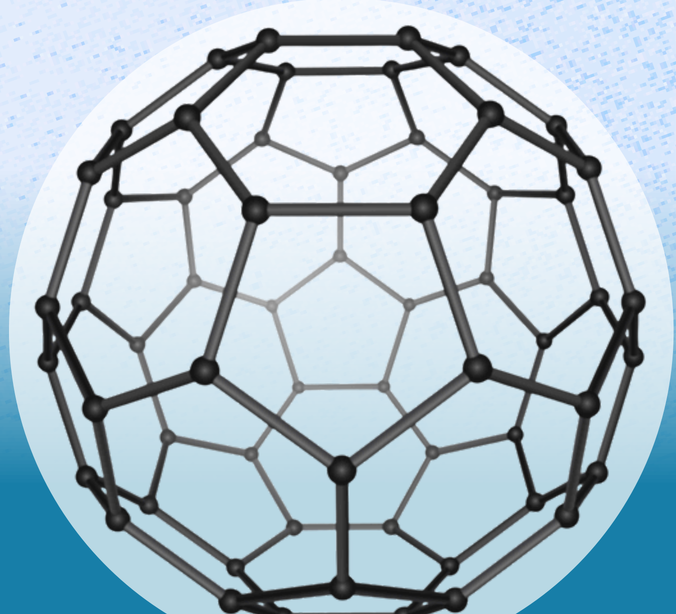
modifying ultrafiltration membrane

SDGS

3 GOOD HEALTH AND WELL-BEING

6 CLEAN WATER AND SANITATION

SIMPLE MOLECULAR STRUCTURE

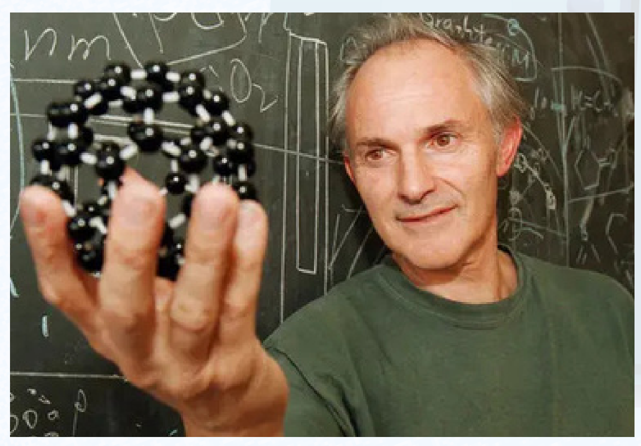


Made up of 60 Carbon atoms

HOW DOES C60(OH)24 FUNCTION IN AN ULTRAFILTRATION MEMBRANE?

1. Hydroxylated buckminsterfullerene is highly hydrophilic due to many (OH)- groups
   - Attracts water molecules to form hydrogen bonds
   - Reduces resistance too water entering and moving through pores
   - Increasing permeability and flux
   - Holds a stable layer of to reject bacteria, colloids, macromolecules etc.
2. C60(OH)24 acts as a pore forming agent
   - Connecting finger-like channels to the surface pores with strong hydration layer
   - Reducing adhesion of proteins and organic foulants
3. Better mechanical stability
   - Rigid buckminsterfullerene core reinforces pore walls
   - Preventing the collapse of finger-like voids under pressure
   - Making it more durable and sustainable

NOBEL PRIZE



awarded to Richard E. Smalley, Robert F. Curl Jr., and Harold W. Kroto

in 1996 for its discovery in Chemistry